MIT SLOAN SPORTS ANALYTICS CONFERENCE presented by ESPN

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A Starting Point for Predicting Team Success in Volleyball

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Introduction

- Analytics has transformed the way professional sports are played at the highest level. For example, the Houston Rockets transformed modern-day NBA offenses by forgoing long two pointers and opting for layups, dunks, and three pointers which were more efficient shots. Teams have poured millions of dollars into their analytics departments, hoping to gain an edge in player evaluation and game strategy through data.
- We seek to study the sports analytic methods used in basketball and apply them to volleyball. In particular, we are interested in creating the volleyball version of the Four Factors equation developed by Oliver Dean. Dean's equation identified how four different areas of the game (EFG%, TO%, OREB%, and FTRate) related to success in basketball and accurately predicted the number of games a team would win in a season. More importantly, Dean determined the importance of each area relative to one another. By developing a similar equation for volleyball, we could determine the volleyball skills most relevant to winning and how influential each skill is.
- By ranking skills in importance, coaches can develop more focused training sessions that will improve the skills most relevant to winning and evaluate players better.

Objectives

- Create a mathematical model that accurately predicts win percentage by conducting regression analysis.
- Determine the most relevant factors for winning sets in volleyball and determine how much each one impacts winning.

Data

- Three seasons of data (2016-2019) on Division 1 and Division 2 Men's Volleyball teams were recorded from VolleyMetrics into an excel spreadsheet. This spreadsheet was converted into a csv file and imported into R. Division 1 and Division 2 teams were grouped together because they often play each other and compete for the same NCAA National Championship. Data on teams with less than twenty sets played was discarded.
- Min-max normalization was performed on data.

Methodology

- We began by coding a step forward regression algorithm in python but transitioned away from this after getting models that did not meet the assumptions of linear regression.
- Instead, we created multiple regression equations with factors that represented each skill and discarded the factors with low coefficient t values. We picked the best model with the lowest RMSE value and the highest R^2.
- Checked linear regression assumptions for chosen model.

Results

- We successfully found a multiple regression model that accurately predicts winning. The three key factors for winning in volleyball are Serve Error % Differential, Serve Ace % Differential, and Attack Efficiency Differential.
- All factors have large coefficient t values, indicating a relationship between the factor and winning exists. Our model also has a MAE value of 0.0284 which means the model is wrong by 2.84% on avg.

```
Residuals:
      Min        1Q    Median        3Q       Max
-0.084080 -0.027636 -0.001424  0.026089  0.095527

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    0.15189    0.01528   9.941  < 2e-16 ***
AcePctDiff     0.21276    0.03082   6.904 3.31e-10 ***
SErrPctDiff   -0.11778    0.01773  -6.642 1.19e-09 ***
AttackEffDiff  0.60355    0.02532  23.833  < 2e-16 ***
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Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' '

Residual standard error: 0.03608 on 111 degrees of freedom
Multiple R-squared:  0.9466,    Adjusted R-squared:  0.9451
F-statistic: 655.5 on 3 and 111 DF,  p-value: < 2.2e-16
```

Table 1: Summary of Model

- All linear regression assumptions are met. The residual plot shows little heteroscedasticity and normal probability plot shows the residuals are normally distributed. All factors appear linear and no significant collinearities exist.

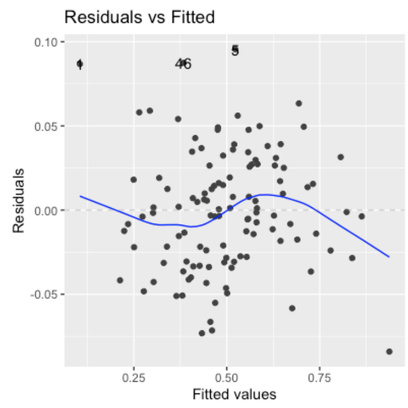

Figure 1: Residual Plot

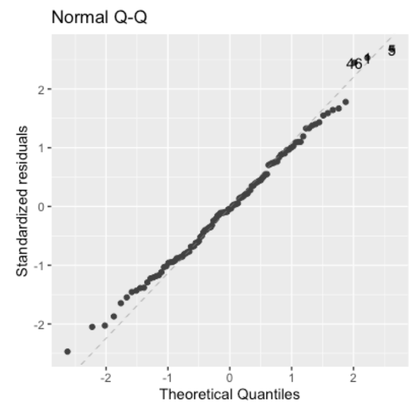

Figure 2: Normal Probability Plot

- Based on the coefficients, the weighting of SE% Diff, SA% Diff, and AttEff Diff is 22.7, 12.6, and 64.7. While we can't say this weighting is definitive because there may be interactions between the dependent variables, the confidence intervals of the coefficients for each variable are distinct ranges, suggesting that AttEff Diff influences winning more than SE% Diff which influences winning more than SA% Diff.

Conclusion

- While we developed a regression model that accurately predicts set win percentage and explains 95% of the variation in wins, we were unable to definitively rank the skills in importance. For one, not all skills were included in the model as there was collinearity between AttEffDiff and the statistics that measured receive and blocking. However, our model suggests that it is more important to take care of the ball and serve in the court rather than maxing out the serve for an ace. It is most important to terminate off sets but too many factors (pass, set, block) influencing AttEffDiff to draw any conclusions.

References

[1] Oliver, D. (2011). *Basketball on paper: rules and tools for performance analysis*. Washington, D.C.: Potomac Books, Inc.

[2] R Core Team (2016). R: A language and environment for statistical computing. R Foundation for Statistical Computing, Vienna, Austria. URL: https://www.R-project.org/